Lecture 2: detailed solutions

Real book exercises — Economics of Organizations and Incentives (MPE Insper)

Solutions to `Listas/lista-reais-lecture-02.pdf`

November 2026

These are detailed solutions to the real book exercises of the list `lista-reais-lectures-01-05.pdf` in `Listas/` (one block of ten per lecture). Within each lecture the items run from easy to hard, in the list's order, and each heading shows the item's number in the lecture, its number in the combined list, its level and its difficulty from 1 (apply a definition) to 5 (graduate level). Statements are reproduced verbatim in the list and are not repeated here. Every solution has the same six parts: (a) what is asked, in one line; (b) the method, with the book section and page and the lecture slide; (c) the solution, every step shown (for matrices, the full dominance comparisons or best-response marking); (d) the economic interpretation; (e) the verification, by the Python check scripts in this folder and against the book's worked answer or the MWG solutions manual where one exists; (f) common mistakes. Books: Tadelis (2013), Muñoz-Garcia & Toro-Gonzalez (2019, 2nd ed.), Gibbons (1992), Mas-Colell, Whinston & Green (1995) (MWG) and Cabral (2017). Notation follows the slides: a game is $G = (I, \{S_i\}_{i\in I}, \{u_i\}_{i\in I})$, s_{-i} is a profile of the others' strategies, S_i^k is the set of player i's strategies that survive k rounds of IESDS with every strictly dominated strategy removed in each round (Lecture 1, slide 29), and $\sigma_i \in \Delta(S_i)$ is a mixed strategy. In bimatrices player 1 picks rows and the first payoff is player 1's; underlined payoffs are best replies.

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Lecture 2 — Static games II: Nash equilibrium and mixed strategies

Item 1 (item 11 of the combined list)**: Tadelis Exercise 5.6** Foundations · difficulty 1/5

Source: Tadelis Exercise 5.6, page 96. The statement is reproduced verbatim in the list (item 11); it is not repeated here.

(a) What is asked. Write Hawk-Dove as a matrix in (v, k, d), then find its pure-strategy Nash equilibria at $v = 10$, $k = 6$, $d = 4$.

(b) Method. Building a matrix from a story (Lecture 1, slides 17–19); pure-strategy Nash equilibrium (Tadelis §5.1, p. 80; Lecture 2, slide 14), found by marking best responses cell by cell (Lecture 2, slide 10; Tadelis §5.1.1, p. 81). The mixed equilibrium mentioned at the end uses the principle of indifference (Lecture 2, slide 44).

(c) Solution. Notation. $v > 0$: value of the resource; k: injury cost when both fight, $k > v/2$; d: display cost when both display, $d < v/2$ (and $d \geq 0$, a cost). $S_1 = S_2 = \{H, D\}$.

(a) The matrix. Each cell from the story:

- both H: they split v and each loses k: $v/2 - k$ each;
- both D: they split v and each pays d: $v/2 - d$ each;
- one H, one D: the hawk gets all of v, the dove leaves with no benefit and no cost, 0.

$1 \setminus 2$	H	D
H	$\frac{v}{2} - k,\ \frac{v}{2} - k$	$v,\ 0$
D	$0,\ v$	$\frac{v}{2} - d,\ \frac{v}{2} - d$

(b) Pure equilibria at $v = 10$, $k = 6$, $d = 4$. Substituting, $v/2 - k = 5 - 6 = -1$ and $v/2 - d = 5 - 4 = 1$. Best responses underlined:

$1 \setminus 2$	H	D
H	$-1,\ -1$	$\underline{10},\ \underline{0}$
D	$\underline{0},\ \underline{10}$	$1,\ 1$

- Player 1 against H: $u_1(D, H) = 0 > -1 = u_1(H, H)$, so $BR_1(H) = \{D\}$.
- Player 1 against D: $u_1(H, D) = 10 > 1 = u_1(D, D)$, so $BR_1(D) = \{H\}$.
- Player 2, by symmetry: $BR_2(H) = \{D\}$, $BR_2(D) = \{H\}$.

Cell by cell:

- (H, H): each would switch to D $(0 > -1)$. Not NE.
- (D, D): each would switch to H $(10 > 1)$. Not NE.
- (H, D): $H \in BR_1(D)$ and $D \in BR_2(H)$. NE, payoffs $(10, 0)$.
- (D, H): by symmetry, NE, payoffs $(0, 10)$.

The same holds for every admissible (v, k, d). Against H, D is strictly better because $0 > v/2 - k \iff k > v/2$. Against D, H is strictly better because $v > v/2 - d \iff d > -v/2$, true for any $d \geq 0$. So the two asymmetric profiles are the only pure equilibria under the book's assumptions; the numbers only make it concrete.

Beyond what is asked: the symmetric mixed equilibrium. Let each play H with probability p. A player is indifferent when

$$\underbrace{p\left(\tfrac{v}{2} - k\right) + (1-p)v}_{EU(H)} = \underbrace{(1-p)\left(\tfrac{v}{2} - d\right)}_{EU(D)} \iff v - p\left(\tfrac{v}{2} + k\right) = \tfrac{v}{2} - d - p\left(\tfrac{v}{2} - d\right) \iff p^* = \frac{v/2 + d}{k + d}.$$

At $(10, 6, 4)$: $p^* = 9/10$, and each expects $EU(D) = (1 - 0.9)(1) = 0.1$.

(d) Interpretation. Hawk-Dove is an anti-coordination game (the same structure as Chicken, Muñoz-Garcia Ch. 3 Ex. 1): each player wants to do the opposite of the other. The two pure equilibria are asymmetric: one animal takes everything, the other backs off. Nothing in the symmetric game says *which* animal is the hawk, which is why the biology literature looks at the symmetric mixed (population) answer: a share $p^* = 0.9$ of hawks, where hawks and doves do equally well. Fighting is frequent because the prize ($v = 10$) is large relative to the injury loss ($k - v/2 = 1$).

(e) Verification. `check_A2.py` builds the matrix from the story with symbols, checks that $BR(H) = D$ and $BR(D) = H$ for 2,000 random admissible (v, k, d), finds by brute force that the pure NE at $(10, 6, 4)$ are exactly (H, D) and (D, H), and checks the general formula for p^*, $p^* = 9/10$ and the payoff $1/10$. Tadelis gives no answer.

(f) Common mistakes.

- Giving the dove a negative payoff when it meets a hawk. The story says it leaves "with no benefits and no costs": 0.
- Forgetting that both H and both D split v: the cells are $v/2 - k$ and $v/2 - d$, not $v - k$ and $-d$.
- Answering "(H, D)" only. The game is symmetric, so (D, H) is an equilibrium too.

Answer. (a) (H, H): $(\frac{v}{2} - k, \frac{v}{2} - k)$; (H, D): $(v, 0)$; (D, H): $(0, v)$; (D, D): $(\frac{v}{2} - d, \frac{v}{2} - d)$. (b) With $v = 10$, $k = 6$, $d = 4$ the matrix is $\begin{pmatrix} -1,-1 & 10,0 \\ 0,10 & 1,1 \end{pmatrix}$; the pure-strategy Nash equilibria are (H, D) and (D, H), with payoffs $(10, 0)$ and $(0, 10)$.

Item 2 (item 12 of the combined list)**: Muñoz-Garcia Ch. 2, Exercise 2** Foundations · difficulty 1/5

Source: Muñoz-Garcia Ch. 2, Exercise 2, page 41. The statement is reproduced verbatim in the list (item 12); it is not repeated here.

(a) What is asked. Find the set of Nash equilibria of the Battle of the Sexes in Fig. 2.4.

(b) Method. Pure-strategy Nash equilibrium by best-response marking (Tadelis §5.1, pp. 80–81; Lecture 2, slides 10 and 14; the slides' own Battle of the Sexes, slide 9). Muñoz-Garcia's Chapter 2 covers pure strategies only, so the book's "set of Nash Equilibria" is the pure set; for completeness the mixed equilibrium is added by the principle of indifference (Tadelis §6.2 and Prop. 6.1, pp. 107–108; Lecture 2, slides 44–50, where the same game is solved in mixed strategies).

(c) Solution. Notation. Husband H (rows), wife W (columns); F = Football, O = Opera.

Best responses, underlined:

$H \backslash W$	F	O
F	$\underline{3},\ \underline{1}$	$0,\ 0$
O	$0,\ 0$	$\underline{1},\ \underline{3}$

- Husband: against F, $3 > 0$, so $BR_H(F) = \{F\}$; against O, $1 > 0$, so $BR_H(O) = \{O\}$.
- Wife: against F, $1 > 0$, so $BR_W(F) = \{F\}$; against O, $3 > 0$, so $BR_W(O) = \{O\}$.

Each player's best reply is to go where the other goes.

Pure equilibria, cell by cell:

- (F, F): $F \in BR_H(F)$ and $F \in BR_W(F)$. NE, payoffs $(3, 1)$.
- (O, O): NE by the same check, payoffs $(1, 3)$.
- (F, O): the husband would switch to O $(1 > 0)$. Not NE.
- (O, F): the husband would switch to F $(3 > 0)$. Not NE.

psNE $= \{(F, F), (O, O)\}$.

The mixed equilibrium. Let $p = \Pr(H \text{ plays } F)$ and $q = \Pr(W \text{ plays } F)$.

- Husband indifferent: $EU_H(F) = 3q$, $EU_H(O) = 1 \cdot (1 - q)$, so $3q = 1 - q$ and $q^* = \frac{1}{4}$.
- Wife indifferent: $EU_W(F) = 1 \cdot p$, $EU_W(O) = 3(1 - p)$, so $p = 3 - 3p$ and $p^* = \frac{3}{4}$.

At $(p^*, q^*) = (\frac{3}{4}, \frac{1}{4})$ each player is indifferent between both pure strategies, so each mix is a best reply to the other (Lecture 2, slide 44). Expected payoffs: $EU_H = 3q^* = \frac{3}{4}$ and $EU_W = p^* = \frac{3}{4}$. The full set of Nash equilibria is

$$\Big\{(F, F),\ (O, O),\ ((\tfrac{3}{4}F, \tfrac{1}{4}O),\ (\tfrac{1}{4}F, \tfrac{3}{4}O))\Big\}.$$

There are no others: $BR_H(q)$ is $p = 1$ for $q > \frac{1}{4}$, $p = 0$ for $q < \frac{1}{4}$ and all of $[0, 1]$ at $q = \frac{1}{4}$; $BR_W(p)$ is $q = 1$ for $p > \frac{3}{4}$, $q = 0$ for $p < \frac{3}{4}$ and $[0, 1]$ at $p = \frac{3}{4}$; the two graphs cross only at $(0, 0)$, $(1, 1)$ and $(\frac{3}{4}, \frac{1}{4})$.

(d) Interpretation. A coordination game with conflict: both want to be together, but they disagree on where. Nash equilibrium identifies the two conventions but cannot say which one the couple will use; that needs something outside the game (a focal point, a habit, who argued last night). In the mixed equilibrium each goes to his or her *own* favourite event with probability $\frac{3}{4}$, they end up apart with probability $\frac{3}{4} \cdot \frac{3}{4} + \frac{1}{4} \cdot \frac{1}{4} = \frac{5}{8}$, and each expects $\frac{3}{4}$, less than the 1 either gets in the worse pure equilibrium.

(e) Verification. `check_A2.py` finds by brute force that the pure equilibria are exactly (F, F) and (O, O), solves both indifference conditions ($q^* = \frac{1}{4}$, $p^* = \frac{3}{4}$), checks the payoffs $\frac{3}{4}$ and the miscoordination probability $\frac{5}{8}$, and finds on a grid of mixed profiles that the mutual best responses are exactly $(0, 0)$, $(1, 1)$ and $(\frac{3}{4}, \frac{1}{4})$. *Comparison with the book's answer* (pp. 41–42): it finds the same best responses and psNE $= \{(F, F), (O, O)\}$; it does not report the mixed equilibrium, which belongs to its Chapter 3. The row label "Housband" in Fig. 2.4 is the book's typo.

(f) Common mistakes.

- Swapping the mixes: the husband's indifference pins down the *wife's* probability ($q^* = \frac{1}{4}$), not his own.
- Reading the answer as "the husband goes to football": an equilibrium is a profile, here (F, F) or (O, O), not one player's choice.
- Claiming the mixed equilibrium is "fairer" and therefore better. Both players get $\frac{3}{4}$, less than in either pure equilibrium.

Answer. Best replies: each goes where the other goes. Pure Nash equilibria: (F, F) with payoffs $(3, 1)$ and (O, O) with payoffs $(1, 3)$ (the book's answer). Including mixed strategies, there is one more: the husband plays F with probability $\frac{3}{4}$, the wife plays F with probability $\frac{1}{4}$, each expecting $\frac{3}{4}$.

Item 3 (item 13 of the combined list)**: Muñoz-Garcia Ch. 3, Exercise 1** Foundations · difficulty 2/5

Source: Muñoz-Garcia Ch. 3, Exercise 1, page 86–87. The statement is reproduced verbatim in the list (item 13); it is not repeated here.

(a) What is asked. For Chicken, find dominated strategies, best responses, the pure and the mixed Nash equilibria, and draw the best-response functions.

(b) Method. Dominance (Lecture 1, slide 23; Tadelis §4.1, pp. 59–61); best responses and pure NE (Tadelis §4.3.1, p. 69, and §5.1, p. 80; Lecture 2, slides 10 and 14); mixed strategies and expected payoffs (Tadelis §6.1, p. 105; Lecture 2, slides 42–43); mixed NE through the principle of indifference (Tadelis §6.2 and Prop. 6.1, pp. 107–108; Lecture 2, slides 44–46): a player mixes only if the other's mix makes him indifferent. The graph in (e) follows the battle-of-the-sexes example (Lecture 2, slides 47–50).

(c) Solution. Notation. St = Straight, Sw = Swerve. Matrix (Fig. 3.1): $(St, St) = (0, 0)$, $(St, Sw) = (3, 1)$, $(Sw, St) = (1, 3)$, $(Sw, Sw) = (2, 2)$. The game is symmetric.

(a) Player 1's rows over columns (St, Sw): $St = (0, 3)$, $Sw = (1, 2)$. Sw is better against St ($1 > 0$), St is better against Sw ($3 > 2$): neither row dominates the other. By symmetry the same holds for player 2. No strictly dominated strategy.

(b) Best responses underlined:

$1 \setminus 2$	St	Sw
St	$0,\ 0$	$\underline{3},\ \underline{1}$
Sw	$\underline{1},\ \underline{3}$	$2,\ 2$

$BR_1(St) = Sw$ (payoff $1 > 0$) and $BR_1(Sw) = St$ (payoff $3 > 2$). By symmetry $BR_2(St) = Sw$ and $BR_2(Sw) = St$. Each player wants the opposite of the other: an anti-coordination game.

(c) A cell is a pure NE when both strategies are best responses to each other. (St, Sw): $St = BR_1(Sw)$ and $Sw = BR_2(St)$, NE. (Sw, St): NE by symmetry. (St, St) and (Sw, Sw) are not, since in each both players would switch. The psNE are (St, Sw) with payoffs $(3, 1)$ and (Sw, St) with payoffs $(1, 3)$.

(d) Mixed equilibrium. Player 1's expected payoffs when player 2 plays St with probability q:

$$EU_1(St) = 0 \cdot q + 3(1 - q) = 3 - 3q, \tag{1}$$
$$EU_1(Sw) = 1 \cdot q + 2(1 - q) = 2 - q. \tag{2}$$

Player 1 is willing to mix only if (1) = (2):

$$3 - 3q = 2 - q \implies 1 = 2q \implies q^* = \tfrac{1}{2}.$$

Player 2's expected payoffs when player 1 plays St with probability p (her payoffs are 0 at (St, St), 3 at (Sw, St), 1 at (St, Sw), 2 at (Sw, Sw)):

$$EU_2(St) = 0 \cdot p + 3(1 - p) = 3 - 3p, \tag{3}$$
$$EU_2(Sw) = 1 \cdot p + 2(1 - p) = 2 - p. \tag{4}$$

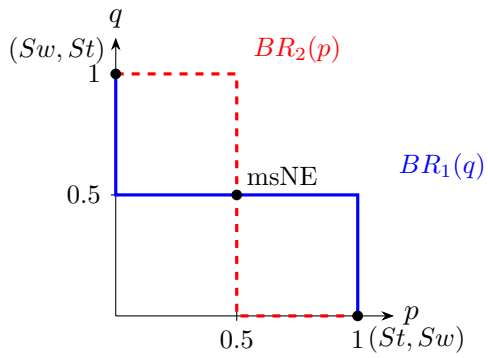
Setting (3) = (4): $3 - 3p = 2 - p$, so $p^* = \frac{1}{2}$. The msNE is

$$\sigma^* = \Big((\tfrac{1}{2}\, St, \tfrac{1}{2}\, Sw),\ (\tfrac{1}{2}\, St, \tfrac{1}{2}\, Sw) \Big),$$

and each player's expected payoff is $EU_1 = 3 - 3 \cdot \frac{1}{2} = 1.5$ (by (1) at $q^* = \frac{1}{2}$).

(e) Best-response functions. From (1)–(2), $EU_1(St) - EU_1(Sw) = 1 - 2q$, so

$$BR_1(q) = \begin{cases} p = 1 & q < \frac{1}{2}, \\ p \in [0,1] & q = \frac{1}{2}, \\ p = 0 & q > \frac{1}{2}, \end{cases} \qquad BR_2(p) = \begin{cases} q = 1 & p < \frac{1}{2}, \\ q \in [0,1] & p = \frac{1}{2}, \\ q = 0 & p > \frac{1}{2}. \end{cases}$$



The two graphs cross three times: at $(p, q) = (1, 0)$, i.e. (St, Sw); at $(0, 1)$, i.e. (Sw, St); and at $(\frac{1}{2}, \frac{1}{2})$, the msNE.

(d) Interpretation. Chicken has the shape of Hawk-Dove (Tadelis 5.6). In the mixed equilibrium each driver goes straight half of the time, so they crash with probability $\frac{1}{4}$, and each expects 1.5, less than the 2 of mutual swerving and less than the 3 of the favourable pure equilibrium. The mix is pinned by the *other* player's indifference, not by one's own payoffs (Lecture 2, slide 50). Raising one's own payoff from swerving would not change one's own mix; it would change the *opponent's*.

(e) Verification. `check_A2.py` checks (a)–(c) by brute force, solves both indifference conditions with sympy ($q^* = p^* = \frac{1}{2}$, payoff $\frac{3}{2}$ for both players, crash probability $\frac{1}{4}$), and finds on a grid of $(p, q) \in \{0, \frac{1}{100}, \ldots, 1\}^2$ that the mutual best responses are exactly $(1, 0)$, $(0, 1)$ and $(\frac{1}{2}, \frac{1}{2})$. *Comparison with the book's answer* (pp. 87–91): it agrees on all five parts (no dominated strategies, the same best responses, the two psNE, msNE with $p = q = \frac{1}{2}$, the same best-response graph, Figs. 3.4–3.6). One typo in the book: on p. 89 the rearranged indifference condition is printed "$3+3q = q+2-2q \Rightarrow q = 1/2$"; from $0q+3(1-q) = 1q+2(1-q)$ it should read $3-3q = q+2-2q$ (as printed, $3 + 3q = 2 - q$ gives $q = -\frac{1}{4}$). The book's result $q = \frac{1}{2}$ is correct.

(f) Common mistakes.

- Solving player 1's indifference for p. Player 1's indifference condition contains only q: it pins down the *opponent's* mix.
- Copying the book's typo line $3 + 3q = q + 2 - 2q$; it gives $q = -\frac{1}{4}$, which is not a probability and should be a warning sign.
- Drawing BR_1 as a function of p on the horizontal axis. BR_1 maps q to p; with p on the horizontal axis its graph has a horizontal segment at $q = \frac{1}{2}$.

Answer. (a) No strictly dominated strategy for either player. (b) $BR_i(St) = Sw$, $BR_i(Sw) = St$. (c) psNE: (St, Sw) and (Sw, St). (d) msNE: each plays Straight with probability $\frac{1}{2}$ ($p^* = q^* = \frac{1}{2}$), expected payoff 1.5 each. (e) $BR_1(q)$: $p = 1$ for $q < \frac{1}{2}$, $p = 0$ for $q > \frac{1}{2}$, any p at $q = \frac{1}{2}$ (symmetric for player 2); the graphs cross at $(1, 0)$, $(0, 1)$ and $(\frac{1}{2}, \frac{1}{2})$.

Item 4 (item 14 of the combined list)**: Gibbons Problem 1.12** Exam · difficulty 2/5

Source: Gibbons Problem 1.12, page 51. The statement is reproduced verbatim in the list (item 14); it is not repeated here.

(a) What is asked. Find the mixed-strategy Nash equilibrium of the 2×2 game with T, B against L, R.

(b) Method. Mixed strategies and expected payoffs (Gibbons §1.3.A, pp. 29–33; Lecture 2, slides 42–43); mixed Nash equilibrium through the principle of indifference (Gibbons §1.3.B, pp. 33–44; Tadelis Prop. 6.1, p. 108; Lecture 2, slides 44–46). First check that there is no pure equilibrium, so any equilibrium must have both players mixing.

(c) Solution. Best responses, underlined:

$1 \setminus 2$	L	R
T	$\underline{2}$, 1	0, $\underline{2}$
B	1, $\underline{2}$	$\underline{3}$, 0

$BR_1(L) = \{T\}$ $(2 > 1)$, $BR_1(R) = \{B\}$ $(3 > 0)$; $BR_2(T) = \{R\}$ $(2 > 1)$, $BR_2(B) = \{L\}$ $(2 > 0)$. No cell has both payoffs underlined: the best responses cycle $(T, L) \to (T, R) \to (B, R) \to (B, L) \to (T, L)$, so there is no pure-strategy Nash equilibrium.

No equilibrium with only one player mixing. If player 2 played a pure strategy, player 1's best reply would be unique and pure, and then player 2's best reply to it would be the other column; so in equilibrium player 2 mixes, and by the same argument so does player 1. Both must be indifferent between their two pure strategies (Lecture 2, slide 44).

Player 2's mix, from player 1's indifference. Let $q = \Pr(L)$.

$$EU_1(T) = 2q + 0(1-q) = 2q, \qquad EU_1(B) = 1 \cdot q + 3(1-q) = 3 - 2q.$$

$$2q = 3 - 2q \implies 4q = 3 \implies q^* = \tfrac{3}{4}.$$

Player 1's mix, from player 2's indifference. Let $p = \Pr(T)$.

$$EU_2(L) = 1 \cdot p + 2(1-p) = 2 - p, \qquad EU_2(R) = 2p + 0(1-p) = 2p.$$

$$2 - p = 2p \implies 3p = 2 \implies p^* = \tfrac{2}{3}.$$

Equilibrium and payoffs.

$$\sigma^* = \Big((\tfrac{2}{3}\,T,\ \tfrac{1}{3}\,B),\ (\tfrac{3}{4}\,L,\ \tfrac{1}{4}\,R) \Big), \qquad u_1^* = 2q^* = \tfrac{3}{2}, \qquad u_2^* = 2p^* = \tfrac{4}{3}.$$

Uniqueness. $EU_1(T) - EU_1(B) = 4q - 3$, so $BR_1(q)$ is $p = 1$ for $q > \frac{3}{4}$, $p = 0$ for $q < \frac{3}{4}$, and $[0, 1]$ at $q = \frac{3}{4}$. $EU_2(L) - EU_2(R) = 2 - 3p$, so $BR_2(p)$ is $q = 1$ for $p < \frac{2}{3}$, $q = 0$ for $p > \frac{2}{3}$, and $[0, 1]$ at $p = \frac{2}{3}$. The graphs cross only at $(p, q) = (\frac{2}{3}, \frac{3}{4})$: it is the unique Nash equilibrium.

(d) Interpretation. This is a Matching-Pennies-like game (Gibbons §1.3.A): player 1 wants to match (T with L, B with R), player 2 wants to mismatch. Each player randomises so as to leave the *other* indifferent, which is why player 1's probabilities come from player 2's payoffs. For instance, raising player 1's payoff at (B, R) from 3 to 4 would leave p^* unchanged and raise q^* to $\frac{4}{5}$ (from $2q = 4 - 3q$).

(e) Verification. `check_A2.py` confirms by brute force that there is no pure equilibrium, solves both indifference conditions with sympy ($q^* = \frac{3}{4}$, $p^* = \frac{2}{3}$), checks the payoffs $\frac{3}{2}$ and $\frac{4}{3}$ and the comparative-statics remark, and finds on a grid of mixed profiles that $(\frac{2}{3}, \frac{3}{4})$ is the only mutual best response. Gibbons gives no answer.

(f) Common mistakes.

- Solving player 1's indifference for p. The condition $EU_1(T) = EU_1(B)$ contains only q, player 2's probability.
- Reporting $(\frac{3}{4}, \frac{2}{3})$ in the wrong order. State which probability belongs to which player and which strategy.
- Stopping at "no pure NE". Nash's theorem (Lecture 2, slide 51) guarantees a mixed one, and the question asks for it.

Answer. No pure-strategy equilibrium. The unique (mixed) Nash equilibrium: player 1 plays T with probability $\frac{2}{3}$ and B with $\frac{1}{3}$; player 2 plays L with probability $\frac{3}{4}$ and R with $\frac{1}{4}$. Expected payoffs $\frac{3}{2}$ for player 1 and $\frac{4}{3}$ for player 2.

Item 5 (item 15 of the combined list)**: Tadelis Exercise 5.8** Exam · difficulty 3/5

Source: Tadelis Exercise 5.8, page 96–97. The statement is reproduced verbatim in the list (item 15); it is not repeated here.

(a) What is asked. Write the n-firm Cournot game in normal form, find its Nash equilibrium, and find the limit of the price as $n \to \infty$.

(b) Method. Normal-form games with continuous strategies (Tadelis §3.1, pp. 46–50; Lecture 1, slide 13) and Nash equilibrium as the crossing of best responses (Tadelis §5.2.3, pp. 87–88; Lecture 2, slides 20–27, with the n-firm case on slides 26–27).

(c) Solution. Book typo. The statement prints "$Q = q_i + \cdots + q_n$". Aggregate production is the sum over all firms, $Q = q_1 + \cdots + q_n$; the subscript i is a typo for 1, and the solution uses $Q = \sum_{j=1}^{n} q_j$.

Notation. $N = \{1, \ldots, n\}$; $q_i \geq 0$ firm i's output; $Q = \sum_j q_j$; $Q_{-i} = Q - q_i$ the rivals' total; $a > c \geq 0$. The book defines $P(Q) = a - Q$ for $Q \leq a$; for $Q > a$ we take $P = 0$ (Gibbons's convention, p. 15). This only matters off equilibrium: there a firm earns $-c\,q_i \leq 0$.

(a) Normal form. Players N; strategies $S_i = [0, \infty)$; payoffs

$$u_i(q_i, q_{-i}) = \max\{0,\ a - q_i - Q_{-i}\}\, q_i - c\, q_i. \tag{1}$$

(b) Best response. Fix Q_{-i}. Raising q_i beyond $a - Q_{-i}$ only adds cost, so consider $q_i \leq a - Q_{-i}$, where $u_i = (a - c - Q_{-i} - q_i)\, q_i$ is strictly concave ($\partial^2 u_i / \partial q_i^2 = -2$). The first-order condition is

$$a - c - Q_{-i} - 2q_i = 0, \tag{2}$$

and with the constraint $q_i \geq 0$,

$$BR_i(Q_{-i}) = \max\left\{0,\ \frac{a - c - Q_{-i}}{2}\right\}. \tag{3}$$

Every firm produces in equilibrium. Suppose some firm has $q_i^* = 0$. By (3) that requires $Q_{-i}^* \geq a - c$, so some rival j produces $q_j^* > 0$ and, by (2) for j, $q_j^* = a - c - Q^*$ (because $Q_{-j}^* + 2q_j^* = Q^* + q_j^*$). But $Q^* \geq Q_{-i}^* \geq a - c$ makes $q_j^* \leq 0$, a contradiction. So every firm is interior and (2) holds for all i.

Solving. Rewrite (2) with $Q_{-i} = Q - q_i$:

$$a - c - (Q - q_i) - 2q_i = 0 \implies q_i = a - c - Q. \tag{4}$$

The right side of (4) is the same for every firm, so all firms produce the same amount, $q_i = q$. Summing (4) over the n firms:

$$Q = n(a - c - Q) \implies (n+1)Q = n(a - c) \implies Q^* = \frac{n(a-c)}{n+1}. \tag{5}$$

Substituting (5) into (4):

$$q_i^* = a - c - \frac{n(a-c)}{n+1} = \frac{a-c}{n+1}, \qquad i = 1, \ldots, n. \tag{6}$$

Since $Q^* < a - c < a$, the price is on the linear part of demand:

$$P^* = a - Q^* = a - \frac{n(a-c)}{n+1} = \frac{a + nc}{n+1}, \tag{7}$$

and each firm earns

$$u_i^* = (P^* - c)\, q_i^* = \Big(\frac{a+nc}{n+1} - c\Big)\frac{a-c}{n+1} = \Big(\frac{a-c}{n+1}\Big)^2.$$

The equilibrium is unique: the argument above showed every NE is interior, and (4)–(6) have one solution. In the slides' notation, $q^m = (a-c)/2$ and (6) reads $q_i^* = 2q^m/(n+1)$; $n=1$ gives the monopoly output q^m and $n=2$ gives $\frac{2}{3}q^m$ (slide 24).

(c) Limit. From (7),

$$P^* = \frac{a+nc}{n+1} = c + \frac{a-c}{n+1} \;\xrightarrow[n\to\infty]{}\; c,$$

and $Q^* \to a - c = q^{cp}$, the competitive output, with each firm's output and profit going to zero. This is the familiar perfectly competitive outcome: price equals marginal cost.

(d) Interpretation. The markup $P^* - c = (a-c)/(n+1)$ falls as $1/(n+1)$. Each firm's output lowers the price for everyone, but a firm counts only the loss on its own units; with more firms its own share is smaller, so it expands more aggressively and the outcome approaches competition (slide 27). Contrast with Bertrand (Lecture 2, slides 29–40), where two firms already price at cost.

(e) Verification. `check_A2.py` solves the n first-order conditions with sympy for $n = 1, \ldots, 6$ and checks (5)–(7) and the profit formula, checks that $P^* - c = (a-c)/(n+1)$ and that the limit is c, and confirms numerically ($a = 100$, $c = 10$, $n = 3$) that no unilateral deviation on a fine grid beats $q_i^* = 22.5$, including the kinked part of demand. No answer in the book.

(f) Common mistakes.

- Treating Q_{-i} as $(n-1)q$ *before* differentiating. Symmetry may be imposed only after the first-order condition is written for a general profile; otherwise firm i seems to move its rivals' output too.
- Assuming the equilibrium is symmetric without proof. The step "$q_i = a - c - Q$ for every i" shows that every equilibrium is symmetric, and the corner argument rules out firms with $q_i = 0$.
- Reading (c) as "each firm's output goes to zero, so the price goes to a". Total output rises to $a - c$, and the price falls to c.

Answer. (a) $N = \{1, \ldots, n\}$, $S_i = [0, \infty)$, $u_i = (a - \sum_j q_j)\, q_i - c\, q_i$ (with price 0 if $\sum_j q_j > a$); the printed "$Q = q_i + \cdots + q_n$" should be $q_1 + \cdots + q_n$. (b) Unique NE $q_i^* = \frac{a-c}{n+1}$, $Q^* = \frac{n(a-c)}{n+1}$, $P^* = \frac{a+nc}{n+1}$, profit $\big(\frac{a-c}{n+1}\big)^2$. (c) $P^* \to c$: the perfectly competitive price (price = marginal cost).

Item 6 (item 16 of the combined list)**: Gibbons Problem 1.6** Exam · difficulty 3/5

Source: Gibbons Problem 1.6, page 49. The statement is reproduced verbatim in the list (item 16); it is not repeated here.

(a) What is asked. Find the Cournot equilibrium with marginal costs c_1, c_2 when both costs are below $a/2$, and when $2c_2 > a + c_1$.

(b) Method. Cournot equilibrium as the crossing of best responses (Gibbons §1.2.A, pp. 14–21; Tadelis §5.2.3, eq. (5.2), p. 87; Lecture 2, slides 20–24), now with a corner: a best response can be $q_i = 0$.

(c) Solution. Notation. $q_i \geq 0$; $P(Q) = a - Q$ for $Q < a$ and 0 for $Q \geq a$ (the referenced model, Gibbons p. 15); firm i's cost $c_i q_i$, $0 \leq c_i < a$; $j \neq i$.

Best responses. For $q_i < a - q_j$ the payoff is

$$u_i(q_i, q_j) = (a - q_i - q_j - c_i)\, q_i, \tag{1}$$

strictly concave in q_i; for $q_i \geq a - q_j$ the price is zero and $u_i = -c_i q_i \leq 0$. The first-order condition of (1) is $a - c_i - q_j - 2q_i = 0$, so

$$BR_i(q_j) = \max\Big\{0,\ \frac{a - c_i - q_j}{2}\Big\}. \tag{2}$$

The corner $BR_i = 0$ occurs exactly when $q_j \geq a - c_i$: the rival alone already pushes the price down to c_i or below.

Interior candidate. If both are interior, (2) gives $2q_1 = a - c_1 - q_2$ and $2q_2 = a - c_2 - q_1$. Substituting $q_2 = (a - c_2 - q_1)/2$ into the first:

$$2q_1 = a - c_1 - \Big(\frac{a - c_2 - q_1}{2}\Big) \implies 4q_1 = 2a - 2c_1 - a + c_2 + q_1 \implies q_1^* = \frac{a - 2c_1 + c_2}{3}, \tag{3}$$

and symmetrically

$$q_2^* = \frac{a - 2c_2 + c_1}{3}. \tag{4}$$

This is a NE exactly when both (3) and (4) are positive.

Case 1: $0 < c_i < a/2$ **for** $i = 1, 2$**.** Then $a - 2c_i > 0$, so $a - 2c_i + c_j > c_j > 0$: (3) and (4) are positive and form a NE. No other NE exists. A corner NE with $q_2 = 0$ would need $q_1 = BR_1(0) = (a - c_1)/2$ and, by (2), $q_1 \geq a - c_2$, that is, $(a - c_1)/2 \geq a - c_2 \iff 2c_2 \geq a + c_1$, impossible because $2c_2 < a < a + c_1$. The same holds for $q_1 = 0$, and $(0, 0)$ fails because $BR_i(0) = (a - c_i)/2 > 0$. The unique NE is

$$q_1^* = \frac{a - 2c_1 + c_2}{3}, \quad q_2^* = \frac{a - 2c_2 + c_1}{3}, \quad Q^* = \frac{2a - c_1 - c_2}{3}, \quad P^* = \frac{a + c_1 + c_2}{3},$$

with profits $u_i^* = (P^* - c_i)\, q_i^* = (q_i^*)^2$, because $P^* - c_i = (a + c_1 + c_2 - 3c_i)/3 = (a - 2c_i + c_j)/3 = q_i^*$.

Case 2: $c_1 < c_2 < a$ **and** $2c_2 > a + c_1$**.** Now (4) is negative, $a - 2c_2 + c_1 < 0$, so there is no interior NE. Try firm 1 as a monopolist:

$$q_1 = BR_1(0) = \frac{a - c_1}{2}, \qquad q_2 = 0.$$

Firm 2's best response to it, by (2):

$$BR_2\Big(\frac{a - c_1}{2}\Big) = \max\Big\{0,\ \frac{a - c_2 - \frac{a - c_1}{2}}{2}\Big\} = \max\Big\{0,\ \frac{a + c_1 - 2c_2}{4}\Big\} = 0,$$

since $a + c_1 - 2c_2 < 0$. So $(\frac{a-c_1}{2}, 0)$ is a NE. It is the only one: the other corner, $q_1 = 0$ and $q_2 = (a - c_2)/2$, would need $(a - c_2)/2 \geq a - c_1$, that is, $2c_1 \geq a + c_2$, but $2c_1 < c_1 + c_2 < a + c_2$; and $(0, 0)$ fails as before. At the NE the price is the monopoly price of firm 1,

$$P^* = a - \frac{a - c_1}{2} = \frac{a + c_1}{2} < c_2,$$

so firm 2 cannot sell any unit profitably; firm 1 earns $\left(\frac{a-c_1}{2}\right)^2$.

(d) Interpretation. With similar costs both firms produce, and the cheaper one produces more: $q_1^* - q_2^* = c_2 - c_1$. When firm 2's cost exceeds firm 1's monopoly price, $c_2 > (a + c_1)/2$, firm 2 is priced out even though it is not shut out by any rule: firm 1 simply behaves as a monopolist and firm 2's best reply is to stay out. The condition $2c_2 > a + c_1$ is exactly "c_2 above firm 1's monopoly price". (Case 1 is sufficient, not necessary, for the interior NE: (3)–(4) are positive whenever $2c_i < a + c_j$ for both firms, even if some $c_i \geq a/2$.)

(e) Verification. `check_A2.py` derives (2)–(4) with sympy, checks $u_i^* = (q_i^*)^2$, and for random cost pairs in each case finds the NE by iterating the exact best responses (2) and confirms it equals the formulas above; it also checks on a fine grid that no unilateral deviation pays, including outputs that push Q above a. No answer in the book. (Kumar's unofficial Gibbons manual, not in `Livros/`, was not consulted.)

(f) Common mistakes.

- Reporting $q_2^* = (a - 2c_2 + c_1)/3$ in the second case. It is negative there; the formula is a Nash equilibrium only when it is non-negative, and the corner of the best response takes over.
- Dropping the $\max\{0, \cdot\}$ from the best response. Without it the second case has no solution at all.
- Checking only that firm 2 does not want to produce. Uniqueness also needs the other corner and $(0, 0)$ to be ruled out, as done above.

Answer. If $0 < c_i < a/2$: unique NE $q_1^* = \frac{a-2c_1+c_2}{3}$, $q_2^* = \frac{a-2c_2+c_1}{3}$, price $\frac{a+c_1+c_2}{3}$, profits $(q_i^*)^2$. If $c_1 < c_2 < a$ and $2c_2 > a + c_1$: unique NE $q_1^* = \frac{a-c_1}{2}$, $q_2^* = 0$; firm 1 is a monopolist at price $\frac{a+c_1}{2} < c_2$.

Item 7 (item 17 of the combined list)**: Gibbons Problem 1.7** Exam · difficulty 3/5

Source: Gibbons Problem 1.7, page 49–50. The statement is reproduced verbatim in the list (item 17); it is not repeated here.

(a) What is asked. Show that in homogeneous-good Bertrand duopoly with unit cost $c < a$ the unique Nash equilibrium is $p_1 = p_2 = c$.

(b) Method. Nash equilibrium (Gibbons §1.1.C, pp. 7–12; Lecture 2, slide 14) in a game with a discontinuous payoff, so there is no first-order condition: existence by checking every deviation, uniqueness by showing that every other profile has a profitable deviation. Gibbons §1.2.B (pp. 21–22) treats the differentiated case; the slides prove this exact result (Lecture 2, slides 29–39, "(c, c) is the unique equilibrium"), and the case split below follows them.

(c) Solution. The game. Players $1, 2$; strategies $p_i \in [0, \infty)$. Demand for firm i is

$$D_i(p_i, p_j) = \begin{cases} a - p_i & p_i < p_j, \\ \frac{1}{2}(a - p_i) & p_i = p_j, \\ 0 & p_i > p_j, \end{cases}$$

read as $\max\{0, \cdot\}$ when $p_i > a$ (no negative sales), and profit is $\pi_i = (p_i - c)\, D_i(p_i, p_j)$, with $0 \le c < a$.

Step 1: (c, c) is a Nash equilibrium. At (c, c) each firm sells $(a - c)/2$ at margin 0 and earns 0. Fix $p_j = c$ and let firm i deviate.

- $p_i > c$: $D_i = 0$, so $\pi_i = 0$. Not better.
- $p_i < c$: firm i takes the whole market, $D_i = a - p_i > 0$ (because $p_i < c < a$), at a negative margin: $\pi_i = (p_i - c)(a - p_i) < 0$. Worse.

No deviation is profitable, so (c, c) is a Nash equilibrium.

Step 2: no other profile is a Nash equilibrium. Take $(p_1, p_2) \neq (c, c)$ and label the firms so that $p_1 \le p_2$ (the game is symmetric). Every such profile falls in exactly one case.

(i) $p_1 < c$. Firm 1 sells a positive amount ($a - p_1$ or $(a - p_1)/2$, both > 0 because $p_1 < c < a$) at a negative margin, so $\pi_1 < 0$. Deviating to any $p_1' > p_2$ gives $D_1 = 0$ and $\pi_1 = 0 > \pi_1$. Not NE.

(ii) $p_1 = c < p_2$. Firm 1 earns 0. Let $\bar{p} = \min\{p_2, a\} > c$ and deviate to $p_1' = (c + \bar{p})/2 \in (c, \bar{p})$: still below p_2, so firm 1 keeps the whole market, $D_1 = a - p_1' > 0$, and $\pi_1 = (p_1' - c)(a - p_1') > 0$. Not NE.

(iii) $c < p_1 < p_2$. Firm 2 sells nothing and earns 0. Let $\bar{p} = \min\{p_1, a\} > c$ and let firm 2 deviate to $p_2' = (c + \bar{p})/2 \in (c, \bar{p})$: it undercuts firm 1, sells $a - p_2' > 0$ at a positive margin, $\pi_2 > 0$. Not NE.

(iv) $c < p_1 = p_2 = p$. If $p \ge a$, demand is 0 and both earn 0; firm 1 deviating to $(c + a)/2$ sells $(a - c)/2 > 0$ at margin $(a - c)/2 > 0$. Not NE. If $c < p < a$, each earns $\pi_1 = \frac{1}{2}(p - c)(a - p) > 0$. Firm 1 undercuts to the midpoint $p' = (p + c)/2$, which is in (c, p), takes the whole market and earns

$$(p' - c)(a - p') = \frac{p - c}{2}\Big(a - p + \frac{p - c}{2}\Big) = \frac{1}{2}(p - c)(a - p) + \frac{(p - c)^2}{4} > \frac{1}{2}(p - c)(a - p).$$

Not NE.

The cases cover every profile with $p_1 \leq p_2$ other than (c, c): $p_1 < c$ is (i); $p_1 = c$ forces $p_2 > c$, case (ii); $p_1 > c$ is (iii) or (iv). With Step 1, (c, c) is the unique Nash equilibrium. ■

(d) Interpretation. With identical products and constant marginal cost, whoever is cheaper takes the whole market, so any price above cost invites a slight undercut that roughly doubles the undercutter's sales. Competition drives price to marginal cost and profits to zero with only two firms: the *Bertrand paradox* (Lecture 2, slide 39), in sharp contrast with Cournot, where two firms keep a markup of $(a - c)/3$ (the n-firm exercise, Tadelis 5.8). The result rests on the continuous price grid, unlimited capacity and homogeneous goods; relaxing any of them (a smallest currency unit, capacity constraints, differentiation as in Gibbons §1.2.B) restores positive margins.

(e) Verification. `check_A2.py` works in exact rational arithmetic with $a = 10$, $c = 2$: it checks that no deviation on a fine price grid (including prices above a) improves on (c, c), and, for 3,000 random profiles $(p_1, p_2) \neq (c, c)$, that the deviation named in the case that applies is strictly profitable. Gibbons gives no answer.

(f) Common mistakes.

- Using a first-order condition. Profit jumps at $p_i = p_j$, so the best response does not exist when $p_j > c$ (one always wants to be slightly below p_j); the argument must be by deviations.
- Forgetting case (ii): $p_1 = c < p_2$ looks "competitive", but firm 1 can raise its price toward p_2 and earn a positive profit.
- Undercutting by "ε" without checking the gain. The midpoint deviation in (iv) shows explicitly that the undercutter earns more than its half of the market.

Answer. At (c, c) both earn 0; raising the price loses all sales and cutting it sells at a loss, so no deviation pays. Any other profile has a profitable deviation: a firm below cost raises its price above the rival's; a firm at cost facing a higher rival price raises its price toward the rival's; a firm priced out undercuts; and at a common price above cost a firm undercuts slightly and nearly doubles its profit. Hence $p_1 = p_2 = c$ is the unique Nash equilibrium.

Item 8 (item 18 of the combined list)**: Tadelis Exercise 6.8** Exam · difficulty 4/5

Source: Tadelis Exercise 6.8, page 125. The statement is reproduced verbatim in the list (item 18); it is not repeated here.

(a) What is asked. Find all pure Nash equilibria of the three-firm entry game, and the symmetric mixed equilibrium.

(b) Method. Pure NE by checking unilateral deviations (Tadelis Def. 5.1, p. 80); mixed NE by the principle of indifference (Tadelis Prop. 6.1, p. 108; Lecture 2, slides 44–46), here with three players.

(c) Solution. Notation. $I = \{1, 2, 3\}$, $S_i = \{E, O\}$ (enter, stay out). With n entrants an entrant's payoff is $\pi(n) = 150/n - 62$; staying out pays 0:

$$\pi(1) = 150 - 62 = 88, \qquad \pi(2) = 75 - 62 = 13, \qquad \pi(3) = 50 - 62 = -12.$$

(a) Pure equilibria. Classify profiles by the number n of entrants.

- $n = 0$: any firm that enters gets $\pi(1) = 88 > 0$. Not NE.
- $n = 1$: an outsider that enters gets $\pi(2) = 13 > 0$. Not NE.
- $n = 2$: each entrant gets $13 > 0$, so it does not want to leave; the outsider would get $\pi(3) = -12 < 0$ by entering, so it stays out. NE.
- $n = 3$: each gets $-12 < 0$ and would rather stay out. Not NE.

The pure NE are the three profiles in which exactly two firms enter: (E, E, O), (E, O, E), (O, E, E), each with payoffs 13 to the entrants and 0 to the outsider.

(b) Symmetric mixed equilibrium. Let each firm enter with probability p. The pure symmetric profiles are not NE (by (a), $p = 0$ and $p = 1$ fail), so $p \in (0, 1)$ and every firm must be indifferent between E and O. Fix firm i. The number of other entrants is 0, 1 or 2 with probabilities $(1-p)^2$, $2p(1-p)$, p^2, so

$$u_i(E, p) = (1-p)^2 \cdot 88 + 2p(1-p) \cdot 13 + p^2 \cdot (-12), \qquad u_i(O, p) = 0. \tag{1}$$

Expanding the first expression in (1):

$$88(1 - 2p + p^2) + 26(p - p^2) - 12p^2 = 88 - 176p + 88p^2 + 26p - 26p^2 - 12p^2 = 88 - 150p + 50p^2.$$

Indifference, $u_i(E, p) = 0$, gives

$$50p^2 - 150p + 88 = 0 \iff 25p^2 - 75p + 44 = 0,$$

$$p = \frac{75 \pm \sqrt{75^2 - 4 \cdot 25 \cdot 44}}{2 \cdot 25} = \frac{75 \pm \sqrt{5625 - 4400}}{50} = \frac{75 \pm 35}{50} \in \left\{\frac{4}{5}, \frac{11}{5}\right\}.$$

Only $p = 4/5$ is a probability. The symmetric mixed NE has each firm entering with probability $p^* = \frac{4}{5}$, and each firm's expected payoff is 0 (the payoff of staying out).

Check: at $p = 0.8$, $(0.2)^2 \cdot 88 + 2(0.8)(0.2) \cdot 13 + (0.8)^2 \cdot (-12) = 3.52 + 4.16 - 7.68 = 0$. Why only interior p can work: $u_i(E, p) = 88 - 150p + 50p^2$ is positive for $p < \frac{4}{5}$ (every firm would enter for sure) and negative for $p \in (\frac{4}{5}, 1]$ (every firm would stay out), so $p^* = \frac{4}{5}$ is the only symmetric equilibrium.

The number of entrants at $p^* = \frac{4}{5}$. It is binomial with $n = 3$: $\Pr(0) = 0.2^3 = 0.008$, $\Pr(1) = 3(0.8)(0.2)^2 = 0.096$, $\Pr(2) = 3(0.8)^2(0.2) = 0.384$, $\Pr(3) = 0.8^3 = 0.512$; the expected number of entrants is $3p^* = 2.4$.

Beyond what is asked: an asymmetric mixed equilibrium. Let firm 1 enter for sure and firms 2 and 3 enter with probability r each. Firm 2 faces one sure entrant and firm 3 mixing, so it is indifferent when $(1-r)\cdot 13 + r\cdot(-12) = 0$, i.e. $r = \frac{13}{25}$; the same holds for firm 3. Firm 1 must prefer entering: $(1-r)^2\cdot 88 + 2r(1-r)\cdot 13 + r^2\cdot(-12) = 88 - 150r + 50r^2 = 88 - 78 + 13.52 = 23.52 > 0$ at $r = 0.52$. So firm 1 entering and firms 2, 3 entering with probability $\frac{13}{25}$ (and the two permutations) are also Nash equilibria.

(d) Interpretation. Entry is a coordination problem over *who* enters: the market supports two firms, not three. The pure equilibria solve it but need the firms to know who stays out. Without such a convention, the symmetric equilibrium has each firm entering with probability 0.8. Then all three enter with probability $0.8^3 = 0.512$ and lose money, and the expected profit of the industry is zero: all the rents of the two-firm market are dissipated by miscoordination.

(e) Verification. `check_A2.py` brute-forces the 2^3 profiles (pure NE are exactly the three two-entrant profiles), solves the indifference condition with sympy (roots $\frac{4}{5}$ and $\frac{11}{5}$), and checks that at $p = \frac{4}{5}$ entering and staying out both pay 0 while $u_i(E, p)$ is positive for $p < \frac{4}{5}$ and negative for $p \in (\frac{4}{5}, 1]$. No answer in the book.

(f) Common mistakes.

- Using a binomial over *three* firms in the indifference condition. Firm i's payoff depends on how many of the *other two* enter; the firm's own choice is what is being compared.
- Keeping the root $p = \frac{11}{5}$, or discarding $\frac{4}{5}$ because the expected payoff is zero. A zero expected payoff is exactly what indifference with staying out requires.
- Concluding that the symmetric equilibrium is the only mixed one. The question restricts attention to it; asymmetric mixed equilibria exist (see above).

Answer. (a) The three profiles with exactly two entrants, (E, E, O), (E, O, E), (O, E, E): each entrant earns $75 - 62 = 13$, and the outsider would earn $50 - 62 = -12$ by entering. (b) Each firm enters with probability $p^* = \frac{4}{5}$ (from $50p^2 - 150p + 88 = 0$); expected payoff 0.

Item 9 (item 19 of the combined list)**: MWG Exercise 8.D.2** Advanced · difficulty 4/5

Source: MWG Exercise 8.D.2, page 263. The statement is reproduced verbatim in the list (item 19); it is not repeated here.

(a) What is asked. Prove that if exactly one strategy profile survives iterated removal of strictly dominated strategies, that profile is a Nash equilibrium.

(b) Method. Iterated deletion of strictly dominated strategies (MWG §8.B, pp. 236–241; Lecture 1, slide 29) and Nash equilibrium (MWG Def. 8.D.1, pp. 246–247; Lecture 2, slide 14). The proof is the slides' Proposition 2 (Lecture 2, slide 17): follow a chain of dominating strategies, along which the payoff against the survivor's profile strictly rises, until it reaches the surviving strategy. The game is finite (MWG §8.B). The same chain works whether the dominators are pure strategies (MWG Def. 8.B.2) or mixed (Def. 8.B.4); both versions are proved.

(c) Solution. Setting. $\Gamma = [I, \{S_i\}, \{u_i\}]$, every S_i finite. $S^0 = S$, and S^{k+1} removes from S^k every strategy strictly dominated given S^k (by a pure strategy in S_i^k, or by a mixed strategy $\sigma_i \in \Delta(S_i^k)$). The process stops after finitely many rounds at S^∞. Hypothesis: $S^\infty = \{s^*\}$.

Claim. For every player i and every $s_i \in S_i$, $u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*)$.

Step 1: one link of the chain. Let $s_i \neq s_i^*$. Then $s_i \notin S_i^\infty$, so s_i was deleted in some round, say round $k+1$: there is a dominator, pure $t_i \in S_i^k$ or mixed $\sigma_i \in \Delta(S_i^k)$, with

$$u_i(\sigma_i, s_{-i}) > u_i(s_i, s_{-i}) \quad \text{for every } s_{-i} \in S_{-i}^k. \tag{1}$$

Since s^* is never deleted, $s_{-i}^* \in S_{-i}^k$, and (1) holds at $s_{-i} = s_{-i}^*$:

$$\sum_{t_i \in S_i^k} \sigma_i(t_i)\, u_i(t_i, s_{-i}^*) = u_i(\sigma_i, s_{-i}^*) > u_i(s_i, s_{-i}^*).$$

A weighted average exceeds $u_i(s_i, s_{-i}^*)$ only if at least one of its terms does, so there is a pure t_i with $\sigma_i(t_i) > 0$ (in the pure case, t_i itself) such that

$$t_i \in S_i^k \quad \text{and} \quad u_i(t_i, s_{-i}^*) > u_i(s_i, s_{-i}^*). \tag{2}$$

Call t_i a *successor* of s_i. Since $t_i \in S_i^k$, it survived at least until the round in which s_i was deleted.

Step 2: the chain. Start at any $s_i^{(0)} = s_i \neq s_i^*$. While the current strategy is not s_i^*, it was deleted at some round, so by Step 1 it has a successor; let $s_i^{(m+1)}$ be a successor of $s_i^{(m)}$. By (2),

$$u_i(s_i^{(0)}, s_{-i}^*) < u_i(s_i^{(1)}, s_{-i}^*) < u_i(s_i^{(2)}, s_{-i}^*) < \cdots$$

The payoffs strictly increase, so no strategy appears twice; S_i is finite, so the chain cannot go on forever. It stops only at a strategy with no successor, that is, at a strategy that is never deleted: s_i^*, the only one in S_i^∞. Hence

$$u_i(s_i, s_{-i}^*) < u_i(s_i^*, s_{-i}^*) \quad \text{for every } s_i \neq s_i^*.$$

Step 3: conclusion. No player gains by a unilateral pure deviation from s^*, so s^* is a Nash equilibrium (in fact a *strict* one). Mixed deviations do not help either, since $u_i(\sigma_i, s_{-i}^*)$ is a weighted average of the pure payoffs, each at most $u_i(s^*)$. ■

Remarks.

- *Uniqueness.* Every Nash equilibrium survives IESDS (Lecture 2, Proposition 1, slide 16), so s^* is also the only pure Nash equilibrium; and in any mixed equilibrium every strategy in the support survives (MWG manual's argument below), so s^* is the only Nash equilibrium of any kind.
- *Finiteness* is used in Step 2: in an infinite game the chain of successors need not end at s^*, and a unique survivor need not be an equilibrium. (Outside the course books: Dufwenberg and Stegeman, *Econometrica* 2002, study this.) Infinite games in the course where IESDS has a unique survivor, such as the Cournot duopoly of MWG 8.B.5, satisfy the conclusion by direct computation.

(d) Interpretation. If common knowledge of rationality alone pins down a single profile, that profile is self-enforcing: no player can do better against what the others are predicted to do. Nash equilibrium is weaker than IESDS in what it assumes about beliefs, and this result says the two agree whenever IESDS is decisive, as in the campaigning game (Tadelis 4.7) or the discrete auction (Tadelis 4.3).

(e) Verification. `check_A2.py` generates random finite games (two players, two to four strategies, integer payoffs) until it has 1,000 with a unique IESDS survivor under pure dominance, and checks that each survivor is a strict Nash equilibrium and the only pure one; it also builds the chain of Step 2 explicitly in each game and checks that payoffs rise along it and that it ends at s_i^*. *Comparison with the MWG solutions manual* (p. 8-9): the manual proves the converse inclusion, that every (mixed) Nash equilibrium survives iterated strict dominance, and concludes that the equilibrium must be the unique survivor. That route needs an equilibrium to exist in the first place, which in a finite game is Nash's theorem (MWG Prop. 8.D.2) but is not stated in the manual's answer. The direct chain argument above needs no existence theorem. The conclusions agree.

(f) Common mistakes.

- Proving the converse (every Nash equilibrium survives IESDS) and stopping. That shows the survivor is the only *candidate*; existence of an equilibrium is a separate step.
- Assuming the strategy that dominated s_i survives to the end. It may itself be deleted later; the chain handles this.
- Comparing payoffs against all s_{-i} instead of against s^*_{-i}. Only the inequality at s^*_{-i} is needed, and it holds because s^*_{-i} is never deleted.

Answer. Let s^* be the unique survivor and $s_i \neq s_i^*$. When s_i was deleted, its dominator (or a pure strategy in its support) t_i was still present and, since s^*_{-i} was also present, $u_i(t_i, s^*_{-i}) > u_i(s_i, s^*_{-i})$. Repeating from t_i gives a chain with strictly rising payoffs against s^*_{-i}; in a finite game it ends at a never-deleted strategy, s_i^*. So $u_i(s_i^*, s^*_{-i}) > u_i(s_i, s^*_{-i})$ for all $s_i \neq s_i^*$ and all i: s^* is a (strict) Nash equilibrium.

Item 10 (item 20 of the combined list)**: MWG Exercise 8.D.6** Advanced · difficulty 5/5

Source: MWG Exercise 8.D.6, page 264. The statement is reproduced verbatim in the list (item 20); it is not repeated here.

(a) What is asked. Prove that every 2×2 game, with arbitrary payoffs, has a mixed-strategy Nash equilibrium.

(b) Method. The hint's route: describe each player's strategy by one probability, write the best-response correspondences, and show they always intersect (MWG §8.D, Def. 8.D.2 and Prop. 8.D.1, pp. 250–251, which is the slides' principle of indifference, Lecture 2, slides 44–46; Tadelis §6.4, pp. 121–122; Gibbons §1.3.B, pp. 41–44). Pure equilibria count: they are degenerate mixed strategies. Nash's theorem (slide 51) gives the result in general; the point here is a direct proof.

(c) Solution. Notation. $p = \sigma_1(a_1) \in [0,1]$, $q = \sigma_2(b_1) \in [0,1]$. Payoffs: $u_1(a_1,b_1) = u$, $u_1(a_1,b_2) = \ell$, $u_1(a_2,b_1) = w$, $u_1(a_2,b_2) = y$; $u_2(a_1,b_1) = v$, $u_2(a_1,b_2) = m$, $u_2(a_2,b_1) = x$, $u_2(a_2,b_2) = z$.

Step 1: the payoff gains are affine. Player 1's gain from a_1 over a_2 when player 2 plays q:

$$D_1(q) = [qu + (1-q)\ell] - [qw + (1-q)y] = q(u-w) + (1-q)(\ell - y). \tag{1}$$

Player 2's gain from b_1 over b_2 when player 1 plays p:

$$D_2(p) = [pv + (1-p)x] - [pm + (1-p)z] = p(v-m) + (1-p)(x-z). \tag{2}$$

Player 1's expected payoff is $u_1(p,q) = u_1(a_2,q) + p\,D_1(q)$, linear in p, so

$$BR_1(q) = \begin{cases} \{1\} & D_1(q) > 0, \\ [0,1] & D_1(q) = 0, \\ \{0\} & D_1(q) < 0, \end{cases} \qquad BR_2(p) = \begin{cases} \{1\} & D_2(p) > 0, \\ [0,1] & D_2(p) = 0, \\ \{0\} & D_2(p) < 0. \end{cases} \tag{3}$$

(p^*, q^*) is a NE iff $p^* \in BR_1(q^*)$ and $q^* \in BR_2(p^*)$.

Step 2: a sign lemma. An affine function D on $[0,1]$ satisfies exactly one of: (i) $D(t) \geq 0$ for all $t \in [0,1]$; (ii) $D(t) \leq 0$ for all t, with $D \not\equiv 0$; or (iii) $D(0)$ and $D(1)$ are nonzero with opposite signs. In case (iii) D has a unique zero $t^* = D(0)/(D(0) - D(1)) \in (0,1)$. (If $D(0)$, $D(1)$ are both ≥ 0 or both ≤ 0, then D, a convex combination of them, keeps that sign on $[0,1]$.)

Step 3: the proof, by cases on D_1.

Case A: $D_1(q) \geq 0$ for all q. Then $p = 1 \in BR_1(q)$ for every q by (3). Let $\hat{q} \in BR_2(1)$; it exists because (3) never gives an empty set. Then $1 \in BR_1(\hat{q})$ and $\hat{q} \in BR_2(1)$: $(1, \hat{q})$ is a NE.

Case B: $D_1(q) \leq 0$ for all q. Symmetrically, $p = 0$ is a best reply to every q, and $(0, \hat{q})$ with $\hat{q} \in BR_2(0)$ is a NE.

Case C: $D_1(0)$ and $D_1(1)$ have opposite signs. Player 1 is indifferent at exactly one $q^* \in (0,1)$. Solving $D_1(q^*) = 0$ from (1):

$$q^* = \frac{y - \ell}{(u-w) + (y-\ell)}. \tag{4}$$

Now look at D_2 and apply the lemma again.

- If $D_2(p) \geq 0$ for all p: $q = 1$ is a best reply to every p; with $\hat{p} \in BR_1(1)$, $(\hat{p}, 1)$ is a NE.

- If $D_2(p) \leq 0$ for all p: with $\hat{p} \in BR_1(0)$, $(\hat{p}, 0)$ is a NE.
- If $D_2(0)$, $D_2(1)$ have opposite signs: player 2 is indifferent at

$$p^* = \frac{z - x}{(v - m) + (z - x)} \in (0, 1). \tag{5}$$

  At (p^*, q^*), $D_1(q^*) = 0$ gives $BR_1(q^*) = [0, 1] \ni p^*$, and $D_2(p^*) = 0$ gives $BR_2(p^*) = [0, 1] \ni q^*$. So (p^*, q^*) is a NE, in strictly mixed strategies.

The cases A, B, C exhaust all possibilities by the lemma, and each produces a NE. Hence every game of this form has a mixed-strategy Nash equilibrium. ■

(d) Interpretation. The argument is the 2×2 version of the fixed-point proof (Tadelis pp. 121–122): each best-response correspondence is a "step" from one side of the unit square to the other, with a vertical (or horizontal) segment where the player is indifferent, and two such steps must cross. Mixing is what makes this work: in pure strategies the steps have gaps (Matching Pennies has no pure NE), and the indifference segment closes the gap. Gibbons's four cases (p. 42) are the sign patterns of $D_1(0) = \ell - y$ and $D_1(1) = u - w$.

(e) Verification. `check_A2.py` checks (1), (2), (4) and (5) with sympy, then draws 20,000 random 2×2 games (half with integer payoffs in $\{-2, \ldots, 2\}$, so that ties and constant gains occur often), runs the case construction above, and confirms with the exact best-response test (3) that the profile it returns is a NE every time. *Comparison with the MWG solutions manual* (pp. 8-15–8-16, where the OCR in `Livros/` prints the number as "8.0.6"): it splits into the four strict-dominance cases ($u > w,\ \ell > y$; $u < w,\ \ell < y$; $v > m,\ x > z$; $v < m,\ x < z$) plus "all other cases", where it gives the mixed equilibrium $\alpha = (y - \ell)/(u + y - \ell - w)$ for player 2 and $\beta = (z - x)/(v + z - x - m)$ for player 1. These are our (4) and (5); the conclusion agrees. Two remarks on the manual: (i) in cases 3 and 4 it says "Player 2 will play his best response" where it means player 1 (a slip: the strategies named are a_1, a_2); (ii) its "all other cases" is not complete as written: if $u = w$ and $\ell = y$ (player 1 indifferent to everything) and player 2 has no strictly dominant strategy, the formula for α is $0/0$ (any α then works, so an equilibrium still exists, but the manual's formula does not exhibit it); the same holds for β when $v = m$ and $x = z$. The case split by the sign of D_1 above has no such gap.

(f) Common mistakes.

- Proving only the generic case. The formulas (4)–(5) divide by $(u - w) + (y - \ell)$ and $(v - m) + (z - x)$, which can be zero; cases A, B and the first two bullets of case C exist to cover exactly those games.
- Forgetting that a pure equilibrium is a mixed one. The claim is existence in $\Delta(S_1) \times \Delta(S_2)$; when one player has a dominant strategy the equilibrium found is pure, and that is enough.
- Citing Nash's theorem as the proof. It is true, but the exercise asks for a direct argument from the best-response correspondences, as the hint says.

Answer. With $p = \Pr(a_1)$, $q = \Pr(b_1)$, player 1's gain from a_1 is $D_1(q) = q(u-w)+(1-q)(\ell-y)$ and player 2's gain from b_1 is $D_2(p) = p(v - m) + (1 - p)(x - z)$, both affine. If $D_1 \geq 0$ (or ≤ 0) on $[0, 1]$, player 1 has a pure strategy that is always a best reply, and pairing it with player 2's best reply to it is a NE; the same if D_2 keeps a sign. Otherwise both change sign, at $q^* = \frac{y-\ell}{(u-w)+(y-\ell)}$ and $p^* = \frac{z-x}{(v-m)+(z-x)}$ in $(0, 1)$, and (p^*, q^*) is a NE because each player is indifferent. So a NE always exists.